

CO4_AT3 : (INDUSTRY PROBLEM SOLVING TASK)

The screenshot shows the SIMATS Online Programming Lab interface. At the top, there's a header with the SIMATS Engineering logo, the title 'SIMATS Online Programming Lab', and the subject 'PYTHON PROGRAMMING'. A user profile for 'Vaibhava Lakshmi K' is visible in the top right. On the left, a sidebar shows 'Q1 Insurance Premium Calculator System' with '50 marks' and '1/1 answered'. The main content area contains the task details:

Scenario / Problem Context:
An insurance company operates across multiple regions and offers a variety of policies including Health, Vehicle, and Life Insurance. Currently, the premium calculation process is handled manually using spreadsheets by different regional offices. This leads to inconsistencies in pricing, difficulty in updating rate tables, and compliance risks during audits. Additionally, regulatory authorities require that every premium quote generated must be stored with full traceability for verification purposes. With increasing customer demand and frequent policy updates, the company needs a scalable and automated system that can dynamically compute premiums using configurable rules and maintain accurate logs for auditing.

Task Description:
Design and implement a Python-based premium calculation engine using modules and packages. The system should load rate tables dynamically from external files and compute premiums based on policy type and input parameters.

Sub-Tasks:

- Create a base class InsurancePolicy with method calculate_premium()
- Derive HealthPolicy, VehiclePolicy, and LifePolicy
- Load rate tables from CSV files dynamically
- Validate inputs and implement exception handling
- Store all generated quotes in a CSV log file with timestamp
- Demonstrate extensibility by adding a new policy (TravelPolicy)

Below the tasks, there's a section for 'UPLOAD PDF FILES * (MAX 2 FILES · 10 MB EACH)'. A file named 'File 1' with the path '192511310RD_q18148_co4_at3_industry_problem_solving_task_1.pdf' is shown with a 'Choose' button and a delete icon.

PROGRAM:

```
import csv
```

```
from datetime import datetime
```

```
rates = {  
    "Health": 0.05,  
    "Vehicle": 0.08,  
    "Life": 0.04,  
    "Travel": 0.06  
}
```

```
class InsurancePolicy:
```

```
    def __init__(self, customer, amount):
```

```
        self.customer = customer
```

```
        self.amount = amount
```

```
    def calculate_premium(self):
```

```
return 0
```

```
class HealthPolicy(InsurancePolicy):  
    def calculate_premium(self):  
        return self.amount * rates["Health"]
```

```
class VehiclePolicy(InsurancePolicy):  
    def calculate_premium(self):  
        return self.amount * rates["Vehicle"]
```

```
class LifePolicy(InsurancePolicy):  
    def calculate_premium(self):  
        return self.amount * rates["Life"]
```

```
class TravelPolicy(InsurancePolicy):  
    def calculate_premium(self):  
        return self.amount * rates["Travel"]
```

```
def save_quote(customer, policy, amount, premium):  
    with open("quotes.csv", "a", newline="") as file:  
        writer = csv.writer(file)  
        writer.writerow([  
            datetime.now().strftime("%Y-%m-%d %H:%M:%S"),  
            customer,  
            policy,  
            amount,
```

premium

)

try:

```
customer = input("Enter customer name: ")
```

```
print("1. Health")
```

```
print("2. Vehicle")
```

```
print("3. Life")
```

```
print("4. Travel")
```

```
choice = int(input("Enter policy choice: "))
```

```
amount = float(input("Enter insured amount: "))
```

```
if amount <= 0:
```

```
    raise ValueError("Amount must be greater than zero")
```

```
if choice == 1:
```

```
    policy = HealthPolicy(customer, amount)
```

```
    policy_name = "Health"
```

```
elif choice == 2:
```

```
    policy = VehiclePolicy(customer, amount)
```

```
    policy_name = "Vehicle"
```

```
elif choice == 3:
```

```
    policy = LifePolicy(customer, amount)
```

```
    policy_name = "Life"
```

```
elif choice == 4:
```

```
    policy = TravelPolicy(customer, amount)
```

```
    policy_name = "Travel"
```

```
else:
```

```
    raise ValueError("Invalid policy choice")
```

```
premium = policy.calculate_premium()
```

```
print("Customer:", customer)
```

```
print("Policy:", policy_name)
```

```
print("Insured Amount:", amount)
```

```
print("Premium:", round(premium, 2))
```

```
save_quote(customer, policy_name, amount, round(premium, 2))
```

```
except ValueError as e:
```

```
    print("Error:", e)
```

OUTPUT:

```
INSURANCE PREMIUM CALCULATOR
-----
Enter customer name: Vaibhav

Available Policies:
1. Health
2. Vehicle
3. Life
4. Travel
Enter policy choice: 1
Enter insured amount: 100000

----- PREMIUM QUOTE -----
Customer: Vaibhav
Policy: Health
Insured Amount: 100000.0
Premium: 5000.0
Quote saved successfully.
```